

Lewis Structures

Valence electrons, multiple bonds, formal charge and resonance

Name _____ Class _____ Date _____

USEFUL INFORMATION

Valence electrons: group number for groups 1–2; group number minus 10 for groups 13–18

Add 1 electron per negative charge; subtract 1 per positive charge

Formal charge = valence e^- – lone-pair e^- – $\frac{1}{2}$ bonding e^-

Draw every lone pair. Hydrogen is always an outer atom.

A. Counting electrons

1. Count the total valence electrons in (a) CH_4 (b) PCl_3 (c) NH_4^+ (d) CO_3^{2-}

B. Drawing structures

2. Draw Lewis structures for (a) H_2S (b) NF_3 (c) CCl_4

3. Draw Lewis structures for (a) formaldehyde, CH_2O (carbon is central) (b) ethene, C_2H_4 (the two carbons are bonded to each other)

4. Draw the Lewis structure of carbon disulfide, CS_2 . How many lone pairs are there in the whole molecule?

5. Draw the Lewis structure of the ammonium ion, NH_4^+ , and calculate the formal charge on nitrogen.

C. Formal charge and resonance

6. Ozone, O_3 , has 18 valence electrons. Draw one Lewis structure and calculate the formal charge on each oxygen atom.

7. Draw the resonance structures of the nitrate ion, NO_3^- . What does resonance tell you about the three N–O bond lengths?

D. Exceptions

8. Explain why BF_3 is an exception to the octet rule. How many electrons surround boron?

E. Challenge

9. Xenon tetrafluoride, XeF_4 , has 36 valence electrons. Draw its Lewis structure. How many electrons surround xenon, and why is this possible?